SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): Better Than Cash Alliance
Solution Maker (SM): Vaka Consulting

1. Challenge Definition

1.1 Refined Challenge Statement

The Indigenous Community of San Miguel Chimalapa, in Oaxaca, Mexico, has decided to implement a forest carbon project focused on protecting and restoring its forest, thereby increasing CO_2 storage and generating carbon credits[1]. Under this project, approximately US $3.25 million per year in potential carbon credit revenues is expected to be directed to a single communal master account managed by the community's governing authority. These funds are intended to be distributed among approximately 4,500 community members across 134,000 hectares of forest; however, current manual cash processes create risks of delays, leakage, and exclusion. At present, there is no robust system in place to ensure transparent and accurate distribution of funds among community members.

Our solution is a blockchain-based payout rail that ensures carbon-credit revenues reach every intended member of the community, providing traceability, auditability, transparency and inclusivity. It is designed to integrate with the community's *usos y costumbres* (customs) governance system, while addressing the main weaknesses of current cash processes: opacity, delays, compliance risks, and exclusion.

1.2 Local Context

Why does this challenge matter?

The challenge addresses opaque, cash-based disbursement of carbon credit revenues in indigenous communities. Currently, there is no system in place to properly distribute funds among community members under *usos y costumbres*, creating risks of delays, leakage, favoritism, and exclusion, especially for women. Lack of traceability erodes trust, threatens conservation efforts, and exposes buyers and investors to audit and ESG risks. A blockchain-based payout rail ensures transparency, accountability, and gender-inclusive distribution, strengthening both climate and social outcomes.

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[1] Carbon credits represent one metric tonne of carbon dioxide equivalent (tCO_2e) reduced, removed, or avoided through a verified climate action. In forest carbon projects, credits are generated by maintaining or increasing forest carbon stocks relative to a defined baseline under an approved carbon standard.

The income generated by the forest carbon project will not only enable the community to strengthen preventive climate actions, but also support economic and social development among its members. The solution is not intended to replace Indigenous governance, but to provide the digital infrastructure needed to uphold it. By codifying customary rules into smart contracts, the solution materially reduces gatekeeper discretion at the point of execution that currently erode community trust and project stability. For the first time, community members can access fully auditable, on-chain records of carbon credit revenues. With this solution, the community is not only protecting a forest; it is de-risking the second largest carbon sink in Mexico by addressing the trust gap in benefit distribution. At a time when many of the world's forests are becoming net carbon sources, Indigenous managed lands remain the planet's most reliable carbon sinks. We provide an auditable payout rail that ensures climate finance reaches the frontline stewards who keep the forest standing.

Setting and key beneficiaries

Setting: Indigenous community of San Miguel Chimalapa, Isthmus of Tehuantepec, Oaxaca, Mexico. Second largest carbon-capture project in Mexico: 134,000 ha of biodiverse tropical forest (Selva Zoque). Governance follows *usos y costumbres* via the General Assembly of *Comuneros*, *Comisariado*, Treasury and Oversight Council.

Key Beneficiaries: Approx. 4,500 landowners (76.8% living below the poverty line). Women (49% of population) and youth are prioritized for inclusion. Community governance bodies benefit from improved trust and transparency.

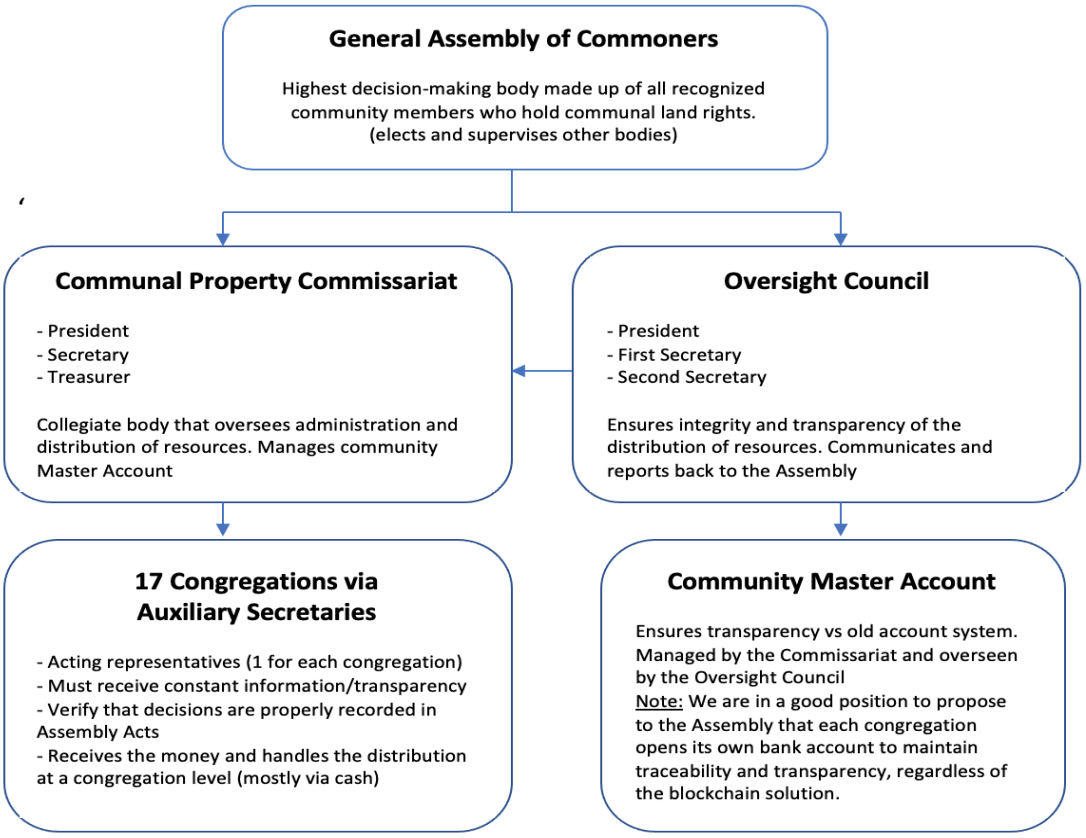


Fig 1: Governing bodies in San Miguel Chimalapa

Why is this challenge important in Mexico and LAC regional context?

Mexico: Indigenous communities manage vast forest areas critical for carbon sequestration. Current manual cash systems perpetuate financial exclusion, gender inequity, and governance friction. The initiative aligns with UNDP priorities and Mexico's National Policy on Financial Inclusion (PNIF) for climate resilience and inclusive development.

LAC Region: Similar governance and payment challenges exist across rural and indigenous communities in the LAC region. Blockchain-based solutions can scale to other carbon projects, renewable energy agreements, and social programs, reducing leakage and improving trust in resource distribution among indigenous and local communities. Countries with similar community governance structures include Peru, Brazil, Colombia, Guatemala and Panama, among others.

Affected community and clean energy sectors, existing efforts, and roles of key stakeholders

Affected Community: San Miguel Chimalapa faces limited financial infrastructure, low digital and financial literacy, and systemic barriers for women under traditional *usos y costumbres* governance. Income depends on forestry/agriculture; carbon credit revenues are a vital new income stream.

Clean Energy/Climate Sector: Carbon Plus manages Improved Forest Management (IFM) and forest restoration under Climate Action Reserve protocols. Activities include thinning, fire prevention, pest control, reforestation, and sustainable timber harvesting.

Existing Efforts: Carbon Plus already operates 100,000 ha of verified carbon projects. Community reinvestment in forest management increased from 15% to 50% of revenue, showing strong conservation commitment.

This blockchain-based solution to distribute carbon credit revenues directly to community members is a groundbreaking initiative—there is currently no other effort like this in the country. This is a unique opportunity to pioneer an innovative, transparent, and equitable model that empowers local communities while driving climate impact.

Key Stakeholders & Roles: Carbon Plus (project developer & Assurance Fund compliance), Community Governance Bodies (approve and oversee fund distribution), UNDP & BTCA (advisory), ClimaFI & Vaka (tech partners), GADES (local ally), Convenience stores (financial agents), Government authorities (SEMARNAT, CONANP and Secretariat of Finance and Public Credit).

1.3 Alignment with UNDP Country Office and National Development Priorities

Explain how the proposed solution aligns with the strategic objectives of the UNDP Country Office and national development frameworks. Does it support existing programs or government priorities (e.g., digital transformation, climate resilience, youth employment, financial inclusion)? Highlight any synergies with UNDP initiatives, government strategies, or donor-funded efforts.

Alignment with UNDP Strategic Objectives & National Development Frameworks

The pilot directly operationalizes several of Mexico´s National Policy on Financial Inclusion 2025-2030 (PNIF) pillars, notably Objective T1: Promoting gender equality and Objective T2: Fostering inclusion of vulnerable groups. Rather than just "promoting" these goals, the solution technically enforces them by hard-coding fair distribution rules into the payout rail.

- By introducing blockchain-based payment rails to channel carbon credit revenues to indigenous landowners, the initiative promotes women's economic empowerment and financial autonomy, per PNIF's mandate to address gender gaps within the financial system.
- This initiative complements the National Development Plan 2025-2030 (PND) and the National Program for Development Finance (PRONAFIDE) strategies—such as Strategy 6.5, which emphasizes financial inclusion in rural and forestry sectors through technology.
- Through interoperable, digital payment systems (QR, USSD/SMS), it aligns with PNIF's enabler objectives to expand coverage, support secure formal payments, and foster trust in digital financial services (Objective H2).

Support for Existing Programs & Government Priorities

The initiative amplifies key government goals outlined in PNIF:

- Digital transformation in underserved areas (leveraging blockchain, USSD/SMS, and QR codes) supports PNIF's push for broad, safe digital financial access, especially in rural zones lacking physical banking infrastructure (70% coverage achieved via tech).
- By delivering fully auditable payouts to 4,500 residents (women and youth included), it reinforces PNIF's target of reaching 83% of the population with at least one financial product by 2030, reducing dependency on informal cash mechanisms.
- Integration of financial education and digital literacy training aligns with PNIF's complementary strategy to its National Financial Education Strategy (ENEF), enhancing capabilities for informed and responsible financial product use.

Synergies with UNDP, Government Strategies & Donor Efforts

This pilot leverages synergies across UNDP, national, and donor initiatives:

- It reflects PNIF's transversal approach, which for the first time integrates gender and vulnerable group inclusion into every stage, thereby aligning with UNDP's focus on inclusive financial governance and accountability.
- Supports the Secretariat of Environment and Natural Resources (SEMARNAT) and the National Commission of Natural Protected Areas (CONANP) conservation goals by aligning climate finance distribution with PNIF's strategies 6.5 & 6.6, which promote formal financing and digital payment solutions to rural, agricultural, and forestry communities (including indigenous groups). This directly protects the "Guardians" of the Selva Zoque by ensuring their financial incentives are not lost to bureaucracy.

- Encourages payment interoperability among local financial providers, echoing PNIF's vision for a robust, diversified, and trusted financial ecosystem supported by 125+ actions and 26 strategies.
- The scalable model integrates with donor-supported efforts (e.g., BTCA) to expand responsible digital payment adoption, forming a replicable blueprint for carbon-credit distribution linked to sustainable development.

1.4 Relevance to UNDP Programmes and Resource Mapping
(Is the pilot building on an existing initiative, or is it a new standalone project? If the project builds on an existing initiative, provide further details. Include scope, scale, timeline and available co-financing or in-kind contributions.

This pilot is not a standalone experiment; it is the critical infrastructure layer for an existing, high-value ecosystem.

Scope & Scale: The initiative builds upon the Carbon Plus "Improved Forest Management" project (CAR 1963), which has already successfully verified the environmental asset. The pilot does not need to raise funds for the payouts themselves; it leverages the US$3.25M in carbon credit revenues generated per cycle as the liquidity source.

The Gap: While the asset is verified, the "Last Mile" of distribution remains manual and risky. This pilot provides the "Compliance Rail" that connects the Comisariado's Master Account to the 4,500 individual landowners.

Co-Financing & Contributions:

- **Carbon Plus:** Providing the US$3.25M revenue stream, community onboarding access, and compliance oversight (Assurance Fund).
- **San Miguel Chimalapa Community:** Providing political ratification via the General Assembly and land stewardship.
- **Better Than Cash Alliance:** providing strategic advisory on responsible digital payments. Ultimately, this system is designed to build trust, streamline operations, and create a sustainable framework for inclusive economic participation. The Alliance will provide $40,000 USD.

To bring this pilot to life and fully realize its potential, additional resources will be essential for implementation. This next step will ensure the solution is tested, refined, and ready to deliver measurable impact for communities and the environment.

1.5 Key Stakeholders & Potential Partners

(Please list all the partners involved in this project. List all relevant parties who will interact with, benefit from, or influence the solution (this may include government agencies, NGOs, community members, or tech partners.))

<u>Relevant Parties</u>

1. **Indigenous Community of San Miguel Chimalapa**

    - Primary beneficiaries and owners of the process. The General Assembly has approved the move to a digital system, ensuring the solution is demand-driven, not top-down. They control the Usos y Costumbres rules that the smart contracts will enforce.
    - They are the rightful landowners of 134,000 hectares of forest, held and stewarded across generations, and are responsible for its protection.
2. **Carbon Plus (Project Developer & Compliance)**
    - Manages the carbon asset (CAR 1963) and the Assurance Fund requirements. They are responsible for lowering the risk profile of the project by ensuring funds reach the right beneficiaries, thereby protecting the project's certification.
3. **Better Than Cash Alliance (Strategic Advisor)**
    - Ensures the pilot aligns with UN Principles for Responsible Digital Payments, specifically regarding gender inclusion and user protection.
4. **ClimaFi & Vaka (Technology Providers)**
    - Responsible for the "Invisible Blockchain" architecture. They provide the Cardano-based backend for auditability and the simplified frontend app that allows unbanked users to access funds via PIN.
5. **Financial Institutions/Agents (Financial Intermediaries) The "Physical Layer."**
    - These partners could provide the local agent network (situated within the community) that allows digital balances to be converted into physical pesos, bridging the gap between the blockchain and the local economy.
6. **Civil Society Organizations and NGOs**
    - Support capacity building and inclusion efforts.
7. **Local Government Authorities**
    - Facilitate regulatory compliance and community engagement.
8. **Regulatory Bodies**
    - Ensure adherence to financial, data and consumer protection standards.
9. **Secretariat of Finance and Public Credit (SHCP)**
    - Interested in participating due to the **development impact on the indigenous community**. Its involvement would include:
        - **Offering financial literacy programs** to strengthen knowledge for informed decision making and responsible usage of digital payments.
        - **Providing financial products and services** tailored to community needs.
        - **Supporting local co-investments** to enhance sustainability and scalability of the initiative.

1.6 Expected Impact

(Outline intended outcomes from the CO's perspective. What would a successful pilot enable - policy change, improved services, empowerment, systems improvement, or scaling innovation?)

<u>Intended Outcomes from the Better Than Cash Alliance and CO Perspective</u>

1. **Policy Change**
    - Provide evidence for integrating **blockchain-based digital payments** into national frameworks for financial inclusion and climate finance.

    - Influence public-sector adoption for transparent social program delivery, demonstrating that "process transparency" can exist without compromising "user privacy."
2. **Improved Services**
    - Replace slow, opaque cash disbursements with **traceable, auditable digital payouts**, reducing payment time from 30–60 days to under 5 days.
    - Ensure **peso-backed redemption mechanisms** for credibility and liquidity.
    - **De-risk the financial flow:** By creating an immutable record of receipts, the project automatically satisfies **Climate Action Reserve (CAR) assurance requirements**, lowering the long-term operational risk of the carbon project.
3. **Community Empowerment**
    - Enable **consent-based transactions** where each landholder authorizes payouts via PIN, reinforcing agency and trust.
    - Technically enforces gender-inclusive entitlements where recognised by community rules (≥45% of payouts to women), bypassing traditional gatekeepers who historically controlled household cash.
    - By securing the financial trust of the 2,630 families protecting Mexico's largest cloud forest, we ensure the forest remains standing.
4. **Systems Improvement**
    - Strengthen governance under *usos y costumbres* by creating a **"Digital Mirror" of the Assembly's will**. This safeguards community leaders from rumors of favoritism and ensures 100% of voted funds reach the intended targets.
    - Provides an evidence-grade audit trail suitable for Assurance Fund verification
5. **Scaling Innovation**
    - Demonstrate a **replicable model** for other CarbonPlus projects (13,000+ landowners across 3 states in Mexico).
    - Enable adaptation for **renewable energy agreements, infrastructure projects**, and **government social programs**, unlocking systemic impact across LAC and beyond.

1.7 Target SDGs and Indicators

(List up to three specific SDG indicators your challenge and proposed solution will directly contribute to. Focus on indicators you will actively address, not indirect ones.)

The project targets directly the following SDGs:

<u>Goal 5 – Gender Equality</u>

Strengthened Indicator Alignment

The pilot directly supports SDG 5.A: *Ensure women's equal rights to economic resources.* By guaranteeing that ≥45% of payouts go directly to women landowners via private, PIN-secured digital wallets, the solution technically enforces gender equity. This bypasses traditional "head of household" gatekeepers, empowering women with independent financial control and strengthening their role in community decision-making.

<u>Goal 10 – Reduced Inequalities</u>

Strengthened Indicator Alignment

Aligned with SDG 10.2: *Empower and promote the social, economic, and political inclusion of all.*

The initiative provides 3,264 indigenous landowners—in a community where 76.8% live below the poverty line—with their first verifiable digital financial identity. By integrating offline-tolerant digital wallets and local agent networks (Yastás), the solution reduces exclusion caused by geographic isolation (50 mins to nearest bank), ensuring equitable benefit-sharing from climate finance.

Goal 13 – Climate Action

Strengthened Indicator Alignment
The project contributes to SDG 13.B: Promote mechanisms for raising capacity for effective climate change-related planning. By securing the financial incentives for the 2,630 families protecting the Selva Zoque, we prevent the "trust erosion" that typically leads to deforestation. The blockchain-based audit trail de-risks the project's compliance profile (Assurance Fund), ensuring that the "monetary promise" of conservation is kept, which is the only way to keep the forest standing long-term.

2. User Mapping & Testing Strategy

2.1 Primary User Persona

Describe the main user or beneficiary of your solution.

Primary User / Beneficiary

Profile: Indigenous landowners and residents of San Miguel Chimalapa, Oaxaca, Mexico. Approximately 6,700 community members, including ~4,500 direct beneficiaries (adults aged 18+).

Context: They live in a rural, heavily forested area governed under *usos y costumbres*, where decisions are made collectively through the General Assembly of Comuneros.

The Trust Gap: Beyond limited financial literacy, this group faces historical "institutional fatigue." They are skeptical of external financial intermediaries and manual cash-handling processes that lack transparency.

Challenges: Long travel distances to access cash, security risks during physical payout, transportation costs and opacity in how funds are calculated and distributed.

Goal: To receive fair, timely, and secure payments from carbon credit sales via a system that is transparent, validated by their own Assembly, that generates accountability and requires no external intermediaries.

2.2 User Story

***The Beneficiary**: "As an indigenous landowner/resident in San Miguel Chimalapa, I want to receive my carbon credit payments directly, transparently, and securely, so that I can support my family, trust the project, and actively participate in protecting our forest."*

- Primary Persona: Josefina (vulnerable rightsholder).
- Profile: 50+ year old indigenous woman, speaks Zoque and Spanish.
- Tech Literacy: Owns a basic feature phone. Limited use of email or Whatsapp.
- Current Pain Point: Travels 50+ minutes to the nearest town (Cabecera Municipal of San Miguel Chimalapa) and depends on a cash handler. Relies on male representatives to collect funds, losing autonomy.
- Her Requirement: "I want to know exactly how much money I'm eligible to receive, and I want to collect it in the most convenient way without the need to travel or handle any cash."

***The Governance Body**: "As a member of the community's governing body, I want to use a reliable system to distribute carbon project revenues fairly within my community, ensuring transparency and traceability."*

- Profile: Male, 30-50s, elected community leaders.
- Tech Literacy: Moderate-high; uses a laptop for minutes and reports, assistants support in tech related activities.
- Current Pain Point: Under constant pressure and expectation to prove transparency and accountability. Fears human-related mistakes that could undermine or destabilize authority and leadership.
- His Requirement: "I need a solid system that proves to the Assembly and every community member that we distribute every peso exactly as it was stipulated".

2.3 Testing Strategy

<u>Timing</u>

Between December 3 and 6, 2025, co-creation workshops were held with community leaders and members of the Community of San Miguel Chimalapa, in Oaxaca, Mexico. These initial participatory sessions were instrumental in identifying specific needs, gathering local perspectives, and providing direct feedback that informed the design, refinement, and implementation of the pilot tool, helping ensure its cultural, operational, and community relevance.

<u>Activities</u>

A second-stage structured testing phase will be carried out with a representative sample of 40 landowners. The three general activities will be:

- Validate usability of key flows:
    - **Registration process**
        - A biometric id will be included
    - **Creation of landowners accounts**

    - **Minting and burning of tokens**
    - **PIN entry for transaction approvals**
    - **Balance and transaction history access**
    - **Communication of the App with the participating banking institution**
    - **Internal operating flow in the participating banking institution**
- Collect feedback on user experience.
    - Implement surveys and interviews (base-line and end-line) to the landowners.
- Analyze the data to identify necessary changes to the app and improve the customer journey.

Note: The onboarding process will be complemented by a financial literacy session offered by the SHCP. The governmental institution to give the financial literacy session is still to be determined by the SHCP.

<u>Goal</u>

Ensure the solution is **intuitive, inclusive, and trusted** by the community before moving to the live pilot phase.

2.4 Feedback Loop Design

How will feedback from users and stakeholders be gathered and integrated?

To ensure the solution is culturally aligned, user-friendly, and compliant with global standards, a structured feedback process will be implemented before the pilot launch. Establishing a baseline requires engaging key stakeholders and end users to validate usability, trust-building, and governance integration. Feedback will be gathered through **structured user testing** with a representative sample of landowners and local agents, focusing on critical flows such as registration, PIN entry, and balance access. Complementary **stakeholder workshops** will involve governance bodies (General Assembly, Comisariado, Oversight Council) to assess cultural fit, while technical partners and BTCA advisors will review adherence to the UN Principles for Responsible Digital Payments. This baseline will inform iterative refinements, ensuring the solution addresses barriers for unbanked and low-literacy users and strengthens security and accessibility features prior to scaling. Finally, all feedback will be integrated into an **end-line analysis**, where data will be synthesized to identify key findings and actionable recommendations for scaling and policy advocacy.

<u>How Feedback Will Be Gathered</u>

- **Structured User Testing (Weeks 9–10):**
    - Engage a representative sample of landowners and local agents.
    - Test usability of registration, PIN entry, and balance access flows.
    - Collect feedback through **interviews, observation, and surveys** during testing sessions (base-line and end-line).
- **Stakeholder Workshops:**
    - Governance bodies (General Assembly, Comisariado, Oversight Council) will provide input on cultural alignment and trust-building.

- Technical partners and BTCA advisors will review compliance with **UN Principles for Responsible Digital Payments**.

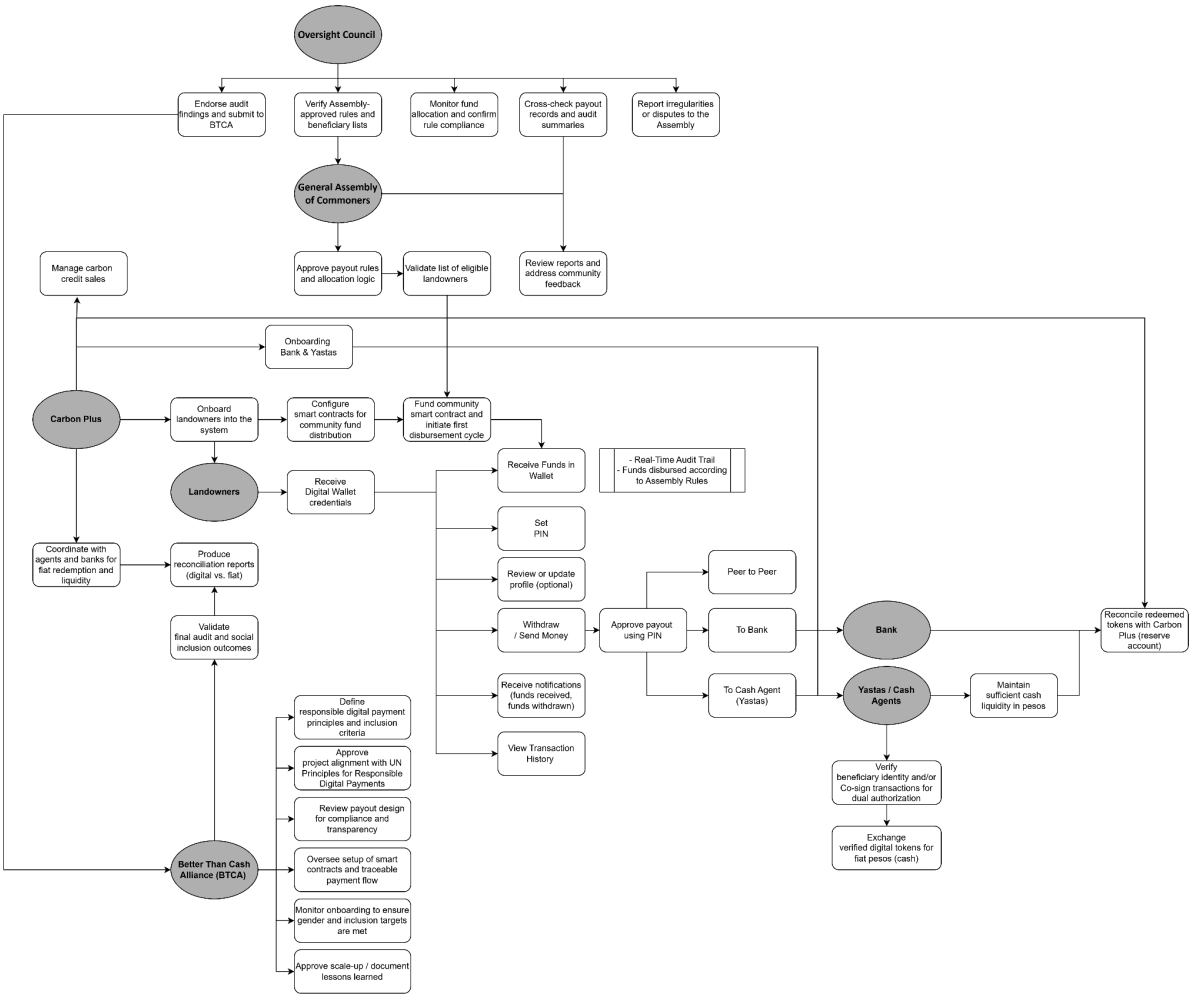
<u>How Feedback Will Be Integrated</u>

- **Iterative Refinement (Weeks 11–12):**
  - Prioritize usability improvements based on user feedback.
  - Adjust UI/UX for clarity and accessibility (e.g., QR codes, USSD flows).
  - Strengthen error handling and security features before pilot launch.
- **Continuous Alignment:**
  - Feedback loops will ensure the solution respects ***usos y costumbres*** governance and addresses barriers for unbanked and low-literacy users.
  - Document lessons learned for scaling and policy advocacy.

3. Solution Overview

3.1 Solution Summary

Briefly describe your proposed blockchain-based solution.



The solution is a blockchain-based compliance and payout rail that ensures carbon-credit revenues reach every intended member of the community, providing traceability, auditability, transparency and inclusivity. It is designed to integrate with the community's *usos y costumbres* (customs) governance system, while addressing the main weaknesses of current cash processes: opacity, delays, compliance risks, and exclusion. It functions as a digital infrastructure layer that lowers the project's "Assurance Fund" risk profile by guaranteeing funds reach the 4,500 beneficiaries.

3.2 Core Features and Functionalities

What are the most important features of your solution?

- **Registration & Identity (Trust)**
  - A new landowner can be registered through the platform (validated against the official Community Census).
  - Registration creates a custodied blockchain wallet managed by the platform (creating a digital financial identity for unbanked users).
  - The landowner sets a PIN code for authorising future transactions.
  - Privacy Shield: Record identity information (e.g., name, gender, ID number) to allow PIN reset, while ensuring on-chain data remains pseudonymous to protect users from extortion risks.
- **Balance & History Access (Transparency)**
  - Each wallet is associated with a **short wallet code**.
  - Landowners can go to a URL and enter their code, or scan a **QR code** to directly access their wallet page.
  - The interface shows **current balance** and **transaction history**, providing real-time transparency into all inflows and outflows.
  - Offline-First Capability: Designed for low-connectivity environments, allowing users to queue transactions.
- **Consent-Based Cash-Out (Security)**
  - Landowners initiate or approve a withdrawal request by entering their PIN.
  - The system builds a transaction from the smart contract holding community funds to the wallet of an approved exchange agent.
  - The transaction requires **multisig authorisation**:
    - The landowner’s wallet signs once PIN is verified.
    - The agent’s wallet co-signs to confirm readiness to deliver cash.
  - Once both signatures are applied, the transaction is executed on-chain (or simulated in the prototype) and the system confirms disbursement (creating the immutable receipt).
- **Audit Trail (Accountability)**
  - Every step (registration, approval, agent co-sign) is recorded on-chain
  - Even in the prototype stage, simulated transactions produce a verifiable log
  - This audit trail ensures full transparency and lays the foundation for real-world verification in the pilot
  - **Compliance Ready:** This audit trail ensures full transparency and lays the foundation for satisfying Climate Action Reserve (CAR) Assurance Fund requirements.

3.3 Tech Stack and Tools

Which technologies and platforms will you use?

Blockchain Layer – Cardano

- Cardano smart contracts will be used for custody and multisignature logic.
- Selected for low transaction costs, scientific rigor, and Proof-of-Stake energy efficiency (Net Zero aligned), which matches the environmental goals of the Carbon Plus project.

Custodied Wallet Service

- Platform-generated wallets securely managed on behalf of landholders.
- Each wallet is linked to cadastral IDs and gender flags, allowing precise entitlement mapping and automated monitoring of the ≥45% female payout target.
- Custody model reduces the technical burden on unbanked or low-digital-literacy users (no private key management required).

PIN-Based Authorisation

- Lightweight PIN entry unlocks signing with the user's wallet.
- Ensures explicit consent for each withdrawal, preventing "head of household" appropriation of funds.

User Interfaces

- Web portal, QR codes, and USSD/SMS channels provide multiple access options.
- Ensures inclusivity: smartphone users (~33%) can scan QR codes, while unbanked or basic-phone users can check balances and approve transactions through USSD or SMS.

3.4 Blockchain Functionality

What blockchain-specific interactions are included (e.g., token minting, smart contracts, on-chain verification)?

The blockchain does not govern the community; it constrains the execution of the community's own decisions.

1. Asset Tokenization (The "Digital Twin")

- **Fiat funds deposited into the Master Account are mirrored by minting equivalent tokens on-chain.** This ensures that the digital ledger always matches the physical bank balance 1:1.

2. Smart Contract Governance (The "Rule Book")

- **Distribution Rules:** Smart contracts execute the specific payout logic approved by the General Assembly (e.g., equal shares vs. land-size shares), removing human discretion from the calculation.
- **Multisig Security:** Large disbursements from the Master Wallet require digital signatures from the **Comisariado (3 keys)**, preventing unauthorized movement of funds.

3. On-Chain Verification (The Audit)

- **Public Explorer Link:** Every transaction generates a hash viewable on Cardanoscan.
- **Metadata Tagging:** Transactions are tagged with non-sensitive metadata (e.g., "Batch #12 Payment") to allow auditors to reconcile bank statements with blockchain records instantly.

Why Blockchain Is Structurally Required for This Use Case

Blockchain is not used for innovation signalling or efficiency gains, but because it is the only infrastructure that remains credible when trust between stakeholders deteriorates.

1. **Immutability under contested governance**
    - The risk is not bugs; it's *accusations*
    - The system must remain credible even if the Comisariado itself is challenged
    - A mutable database fails precisely when trust breaks
2. **Shared truth across adversarial stakeholders**
    - Community members, Carbon Plus, CAR auditors, BTCA, and buyers do not trust the same institution
    - Blockchain provides a neutral, third-party verifiable record without creating a new power centre
3. **Evidence-grade auditability**
    - CAR Assurance Fund and ESG buyers require post-hoc, tamper-proof evidence, not internal logs
    - Blockchain produces *verifiable proof*, not attestations

4. Weekly Workplan

Briefly describe your team's focus during each stage of the program. (You may use the workplan template from your hackathon roadmap.)

Timeline (Week 1, 2, 3..)	**Focus / Category** (Blockchain Interaction, User Interface, Stakeholder Testing, SDG Integration Demo Readiness, etc.)	**Baseline Metric** (One meaningful blockchain function implemented (e.g., token minting, VC issuance, on-chain

		hash; etc.)
Weeks 1–2	**Design & Setup (Establish foundation)** ● Finalize prototype scope and data model (wallets, users, transactions) ● Configure development environment and blockchain connection (Blockfrost / smart contract framework) ● Draft smart contract logic aligned with *Usos y Costumbres* rules (e.g., equal split vs. land-size split).	Development environment operational; initial smart contract compiled
Weeks 3–4	**User Registration & PIN Flow (Enable landholder onboarding)** ● Implement backend service to register landholders ● Create wallet generation and custody service ● Add PIN setup and verification for transaction approvals ● Minimal admin interface for registering users	First landholder successfully registered with wallet + PIN
Weeks 5–6	**Balance & Transaction History Interface (Provide transparency)** ● Build web page to display wallet balance and history ● Generate short wallet codes and QR codes for direct access ● Integrate blockchain API for real transaction lookups	Landholder can view balance and history using short code/QR
Weeks 7–8	**Multisig Transaction Approval (Prototype Cash-Out)** ● Implement 2-of-2 signing flow (landholder wallet and agent wallet) ● Build simple interface for agent co-signing ● Simulate “cash payout” once transaction confirmed ● Log approved transactions for audit	First successful multisig transaction executed with PIN + agent co-signature
Weeks 9–10	**User Testing & Feedback (Validate usability)** ● Conduct structured testing with sample users and agents ● Collect feedback on registration, PIN entry, and balance access flows ● Identify and prioritise refinements	Prototype tested with sample users; feedback captured

Weeks 11–12	**Refinement & Demo Readiness (Prepare for pilot)** ● Refine UI/UX based on user testing ● Fix bugs and strengthen error handling ● Prepare demonstration materials and walkthrough for stakeholders	Prototype stable and demonstrated end-to-end

5. Final Deliverables

5.1 Solution Demo (PoC)

A working or visual prototype that demonstrates the core idea. (To be scheduled and presented virtually to CO team and UNDP/Partner FPs in a weekly coordination meeting)

DONE (recorded by UNDP/Cardano) - Demo Day - [UNDP Better Than Cash Alliance & ClimaFi]

5.2 Testing & Feedback Documentation

Summary of user testing activities, challenges, and feedback.

A co-creation session with the community leaders took place in Juchitan, Oaxaca, Mexico, on December 3rd, 2025. During the session, the mockup of the blockchain-based digital payments solution was presented to the leaders, and they were guided through the different steps to make a transaction.

Key Feedback & Challenges:

- **The "Seed Phrase" Barrier:** Leaders flagged that the standard blockchain onboarding (writing down a 12-word seed phrase) was a critical failure point. They expressed high anxiety that older landowners would lose their phrases and thus their funds.
- **Terminology Confusion:** Terms like "Minting" and "Burning" tokens were confusing and sounded destructive. Participants preferred language that mirrored their reality, such as "Assigning Balance" and "Cashing Out".
- **Trust vs. Privacy:** While leaders appreciated the *internal* transparency of the dashboard, they were concerned about *external* privacy. They requested that the public block explorer not reveal the names of beneficiaries to the general public to prevent extortion risks.
- **Offline Anxiety:** A major concern was connectivity. Users asked, "What happens if the internet cuts out while I am at the Agent?" needing visual reassurance that the transaction was queued.

The session recording and feedback are available here.

5.3 Feedback Integration Summary

Description of what changes were made based on testing and feedback.

Validation of Custodial Architecture: The testing successfully validated our initial architectural decision to use a Platform-Managed Custodial Wallet rather than self-custody. We maintained this design as it proved to be the only viable method to ensure inclusion for older, non-technical landowners without compromising security.

Localization of Interface Language: We refined the frontend copy to align with the community's specific dialect and financial mental models. Technical placeholders were replaced with culturally validated terms (e.g., using "Saldo" and "Retiro") to ensure immediate comprehension by the General Assembly.

Reinforced Privacy Logic: We finalized the separation of on-chain and off-chain data. The feedback confirmed that our "Privacy Shield" design, which restricts real-name mapping to the secure Admin Dashboard while keeping the public ledger pseudonymous, is the correct approach to mitigate local security risks.

Enhanced Connectivity Feedback: Building on the existing "Offline-First" capability, we added specific UI indicators (e.g., a "Queued/Pending" yellow state) to provide immediate visual reassurance to users when transacting in low-signal areas.

5.4 Pilot-Ready Solution

A stable and functional version of the solution, ready for pilot deployment.

The solution has graduated from the concept phase to a functional Minimum Viable Product (MVP). The core infrastructure, including the Cardano smart contracts, the PIN-based multisig logic, and the web-based "Transparency Dashboard", has been built and successfully demonstrated during the co-creation sessions with community leaders in Juchitán (December 2025).

Current Capabilities:

- **Smart Contracts Deployed:** The governance logic (distribution rules) and security logic (multisig approvals) are coded and operational on the Cardano testnet.
- **User Interfaces Validated:** The critical "Landowner Flow" (login via Wallet Code, check balance, approve via PIN) has been tested with representative users, confirming that non-technical users can successfully authorize transactions.
- **Audit Trail Generation:** The system successfully generates immutable on-chain logs for every simulated transaction, proving the technical feasibility of the "Real-Time Audit" value proposition.

Gap to Live Pilot: The technology is ready. The remaining requirements to move from this functional MVP to the live pilot are purely operational: (1) Final deployment to Mainnet, (2) Onboarding of the liquidity agents (Yastás), and (3) Securing the operational budget for the field team. We are confident the solution can support a live pilot immediately upon funding approval.

5.5 Confirmed Pilot Partnership(s)

Documented confirmation of at least one partner committed to implementation.

1. The **Statement of Intent** signed between Carbon Plus and the Better Than Cash Alliance can be found here.
2. San Miguel de Chimalapa has confirmed its interest in participating in the pilot. A paragraph addressing the pilot has been incorporated into the **Internal Regulation** to be signed between Carbon Plus and San Miguel de Chimalapa; however, this documentation will likely not be available for a couple of months, as it must first be reviewed and formally approved during the community´s General Assembly.
3. The **Secretariat of Finance and Public Credit** (SHCP) has shown strong interest in joining the pilot, recognizing its potential to drive development within indigenous communities. An initial meeting was held in December 2025, and following that discussion, the Head of Development Banking requested a second meeting, scheduled for early February 2026. Please refer to the related email correspondence here for further details.

5.6 Stakeholder Alignment Matrix

Defined roles, responsibilities, and a coordination plan with key partners.

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement (Lead / Support / Advisory)	Engagement Modality (e.g. MOU, Contract)	Focal Point / Contact Person
Carbon Plus	Carbon project developer working in partnership with the community. Responsible for selling carbon credits and disbursing the community's share of revenues.	Provide operational requirements; coordinate with community governance; enable access to payment flow/process data; support rollout and troubleshooting; ensure continuity with carbon-credit programme rules	Lead	Implementation agreement / LOA (pilot collaboration)	Javier/Berenice Project Director

San Miguel Chimalapa Community Governance (Asamblea de Comuneros, Comisariado, Consejo de Vigilancia)	Legitimation, governance, oversight, and community adoption	Validate beneficiary roster and eligibility rules; approve governance procedures (consent/PIN, dispute resolution); overseegen transparency and accountability; community communications	Lead	Community agreement / Acta de Asamblea + working protocol	TBC – Comisariado President / Assembly Secretary
Better Than Cash Alliance (Challenge Owner)	Responsible digital payments, inclusion, learning, and coordination support	Provide responsible digital payments guidance; inclusion/gender safeguards; partner coordination; documentation and learning for scale/replication	Lead + Advisory	In-kind technical assistance (internal workplan)	Maria Antonieta Paz as focal point; Gisela Davico and Diego Mendiburu as support team (BTCA)
UNDP Mexico Country Office	UNDP programme alignment + government interface	Ensure alignment with UNDP portfolio and risk management; convene/coordinate with government counterparts; support safeguards (data protection, inclusion); facilitate scale pathways	Support	Internal UNDP coordination + partner letters as needed	TBC – Programme Specialist Rosamaria Velez as SFH Focal Point and BIOFIN Team

UNDP SDG Blockchain Accelerator (Secretariat / Technical Team)	Technical and delivery oversight during pilot-readiness + knowledge	Provide accelerator technical guidance; quality assurance on blockchain approach; support documentation for Executive Board/knowledge products; connect to UNDP practice networks	Advisory	Internal UNDP support	Burçu Mavis
Vaka Consulting (Solution Maker)	Solution design, build, testing, deployment support	Develop and maintain pilot version; configure smart contracts; conduct security testing; optimize user journey; support deployment;	Lead	Contract	Matt Roberts-Davies & Phil Lewis Technical Lead / Product Owner
Digital Wallet / Payment Service Provider (Bank/Fintech/EMI) – to be confirmed	Payments rail + wallets + compliance	Provide wallets/accounts; KYC/tiers appropriate for context; transaction processing; settlement and reconciliation; service-level support; compliance with AML/CFT and consumer protection	Lead / Support (depending on model)	Commercial contract + SLA	TBC – Partnerships Lead
CICO Networks (local merchants / convenience stores) – to be confirmed	Last-mile cash-out and liquidity	Provide cash-out services; maintain liquidity; follow agent procedures; support onboarding and basic user support; record/confirm	Support	Agent agreements (per agent/aggregator)	TBC – Agent Network Manager

		cash-out steps as required			
Secretariat of Finance and Public Credit (SHCP)	Public-sector enabling partner (inclusion + literacy + potential co-investment)	Support financial literacy; advise on enabling conditions for inclusion; explore linkages to national financial inclusion priorities; consider co-investment for scale	Advisory / Support	MOU / Letter of collaboration (non-financial or co-financing)	Rogelio Rivero Márquez, Head of Development Banking Abraham Mendoza, Director for Banking Financial Services Leslye Lenia Mejía Martínez, Legal Director for Legislation on Financial System for Development

5.7 Budget Proposal (Pilot)

Provide a budget breakdown for pilot implementation. Include estimated cost, unit, quantity, and funding source.

All costs are structured to reflect the pilot's operational needs and strategic priorities, ensuring alignment with UNDP's principles for responsible digital payments and inclusive financial ecosystems.

Budget Line Item	Unit / Description	Quantity	Unit Cost (USD)	Total Cost (USD)	Funding Source (Confirmed / Pending)
Developer Contract (App and Dashboard) - Design & Architecture: $4,500 (monthly) - Client/Project Manager: $4,500 (monthly)	Monthly retainer	6 months	$19,000	114,000	Pending (Grant application)

- Technical Lead/Blockchain Development: $3,000 (monthly) - Software Engineering: $8,000 (monthly)					
Project Manager	Monthly retainer (delivery coordination in the field; probably NIPSA contract)	4	3,000	12,000	Pending (Grant application)
Pilot evaluation a) Onboarding of rightsholders b) Surveys and interviews c) Monitoring and analysis d) Report on Findings e) Report on Recommendations for Scaling		1	25,000	25,000	50% covered by BTCA budget, the rest by grant application
Internet Routers/starlink	Routers for pilot sites/support points	8 units	200	1,600	Pending (Grant)
Cloud Services	Hosting, monitoring, backups, logs	6 months	500	3,000	Pending (Grant)

Comm material (Brochures, Videos, Radio Content)	Brochures + radio content + short explainer video + community comms	1 package	5000	5,000	BTCA budget
Missions to San Miguel Chimalapa (3 people)	Flights/per diem/local transport (implementation + oversight)	2 missions	1,500	3,000	BTCA Budget
Contingency				10,000	Grant application
TOTAL ESTIMATED COST				**173,600**	

5.8 Impact Measurement Framework

5.8.1 Contribution to relevant SDGs and indicators

It aligns with SDGs 5 (Gender Equality), 10 (Reduced Inequalities), and 13 (Climate Action), ensuring transparency, inclusion, and accountability through blockchain-enabled digital payments.

- Rationale for SDG 5 – Gender Equality

  Ensuring women landowners have equal access to financial resources promotes gender equality and empowers vulnerable groups.

- Rationale for SDG 10 – Reduced Inequalities

  Financial inclusion initiatives reduce economic disparities by providing marginalized communities with access to secure and transparent payment systems.

- Rationale for SDG 13 – Climate Action

  Blockchain-enabled payments support climate resilience by ensuring funds for forest management and carbon sequestration reach beneficiaries efficiently.

5.8.2 Key Performance Indicators (KPIs)

This Monitoring & Evaluation (M&E) Framework provides a comprehensive approach to tracking progress and impact of the SDG Blockchain Accelerator Pilot in San Miguel Chimalapa.

The pilot will be considered unsuccessful if it fails to demonstrably reduce disputes, improve payout timeliness, or maintain community legitimacy. In such a case, the blockchain ledger will still serve as a forensic record to diagnose failure modes and inform future governance or payment design.

Monitoring & Evaluation Indicators

Indicator	Baseline	Target	Data Source	Frequency	Responsible Party
% of beneficiaries registered with digital wallets	0%	>90%	Platform records	Quarterly	Digital Payments App Developers
Average payout time (days)	30-60 days	<1 day	Transaction logs	Monthly	Better Than Cash Alliance
% of payouts processed via blockchain	0%	100%	Blockchain explorer	Monthly	Digital Payments App Developers
% of women beneficiaries receiving direct payments	Unknown	≥45%	Gender-disaggregated data	Quarterly	Carbon Plus
System uptime (%)	N/A	>99%	System logs	Monthly	Digital Payments App Developers

Blockchain-Related Indicators

Indicator	Baseline	Target	Data Source	Frequency	Responsible Party
% of payouts processed via blockchain	0%	100%	Blockchain explorer	Monthly	Developers
% of transactions recorded on-chain	0%	100%	Blockchain logs	Monthly	Developers

System uptime (%)	N/A	>99%	System logs	Monthly	Developers
Number of successful PIN-based approvals	0	All withdrawals	Platform records	Monthly	Developers

Gender and Household Indicators

Indicator	Baseline	Target	Data Source	Frequency	Responsible Party
% of women registered with digital wallets independently	0%	65%	Registration logs	Monthly	Better Than Cash Alliance
% of women completing transactions without assistance	0%	65%	App analytics	Monthly	Developers
% of households where women initiate digital payments	40%	49%	Household survey	Monthly	Better Than Cash Alliance
% of women reporting autonomy in financial decisions	0%	65%	Endline survey	Monthly	Better Than Cash Alliance
% of households reporting improved budgeting practices	0%	75%	Endline survey	Monthly	Better Than Cash Alliance

Survey Questions for Indicator Generation

The following questions are designed to capture data for all indicators, including digital payments adoption, usage patterns, gender empowerment, household decision-making, and perceptions.

Digital Payments Adoption

a) Do you currently use digital payment methods? (Yes/No)
b) What type of phone do you use? (Smartphone/Basic)
c) Do you have access to the internet? (Yes/No)

Usage Patterns

d) How often do you use digital payments? (Daily/Weekly/Monthly)
e) What types of digital payment services do you use? (Wallet, Bank, Card)

Gender Empowerment

f) Who in your household decides on financial matters? (Self/Partner/Joint)
g) Do you feel confident making financial decisions independently? (Yes/No)

Household Decision-Making

h) Who decides on major household expenses? (Self/Partner/Joint)
i) Do you participate in decisions about reinvestment of income? (Yes/No)

Perceptions

j) How do you rate digital payments in terms of security? (High/Medium/Low)
k) What are the main benefits you perceive from using digital payments?

5.8.3 Monitoring & Reporting Plan

Frequency: Baseline, monthly monitoring, and end-line.
Responsible Parties: Developers, App Administrator, and Better Than Cash Alliance
Format: Digital payments platform, surveys, and interviews
Tools: Digital dashboard/console, digital surveys ,and personal interviews.

At baseline, the pilot will establish starting conditions and confirm operational readiness by verifying how many beneficiaries are validated and successfully onboarded, and the share that can authenticate and access their balance. It will also document the current cash-based process (time and costs, steps, typical delays, and leakage risks) to create a clear “before” benchmark. Finally, it will assess inclusion conditions, including women’s onboarding rates and barriers for low literacy or limited connectivity, alongside initial levels of trust and confidence in the new digital payment system.

During monthly monitoring, the team will track whether the pilot is operating reliably and inclusively by reviewing dashboard data on transaction completion, delays and failures, user activity, and any exceptions requiring manual intervention. In parallel, short pulse surveys and targeted interviews will capture user experience, highlighting pain points, trust levels, and practical barriers. Findings will be used to update the risk and action logs, prioritize fixes with the developers and app administrator, and produce a brief monthly note summarizing performance, issues, corrective actions, and next steps.

At the end-line, the pilot will assess results and scale readiness by comparing outcomes against the baseline—measuring changes in payout speed, cost, and reliability, as well as the completeness of reconciliation and overall transaction success. It will also consolidate evidence on user experience and inclusion (including women’s adoption and satisfaction), documenting what worked, what failed, and how issues were resolved. The end-line will culminate in a final evaluation and a practical set of recommendations for scaling, including governance requirements, operational improvements, and an implementation roadmap.

5.8.4 Risks Assessment & Mitigation Plan

The primary worst-case risk is not technical failure, but deliberate favouritism or misappropriation of funds by individuals involved in cash handling or beneficiary selection, which could undermine trust, trigger community conflict, and jeopardise the carbon project’s certification.

Risks	***Mitigation***
Governance & legitimacy: Community governance bodies do not fully endorse rules (eligibility, dispute handling, data use), undermining adoption	Co-design key rules, formalize approval through agreed community processes (e.g., assembly acts); transparent communications; designate community focal points
Beneficiary registry integrity: Incorrect/duplicate beneficiary records lead to wrong payouts or disputes	Multi-step validation; unique identifiers; audit trail; pre-payout verification window; exception workflow for corrections
Inclusion/gender gap: Women and other vulnerable groups are onboarded at lower rates or face higher drop-off	Targeted onboarding sessions for women; female facilitators where possible; simple UX; offline support; monthly sex-disaggregated monitoring and corrective actions
Low digital literacy: Users struggle with authentication and basic app use, reducing uptake	Simple step-by-step materials; training-of-trainers; short refresher sessions before payout cycles
Connectivity constraints: Weak coverage disrupts onboarding and transaction confirmations	Use routers in pilot points; schedule transactions around connectivity windows; provide offline fallback procedures for confirmations
Platform reliability: Bugs or downtime delay payouts and erode trust	Design the platform with a minimal, well-tested architecture and deploy it on reliable cloud infrastructure with basic monitoring and alerting in place. Conduct staged testing and dry-run payout cycles before live use, and maintain clear fallback procedures (including manual payout approval) to ensure payments can continue if the system is temporarily unavailable.
Transaction errors/exceptions: Failed/duplicate transactions create confusion and reputational risk	Exception handling playbook; clear user messaging; reconciliation checks; manual review workflow; monthly exception reporting
Data Protection and Consent: Personal data is mishandled, or consent is unclear, causing harm or non-compliance	Minimise the collection of personal data to only what is strictly necessary, clearly document and communicate consent at the point of registration, and store all sensitive information securely with access controls and audit logs. Regularly review data-handling practices to ensure alignment with applicable data protection principles and community governance expectations.
Fraud: Users are pressured to share PIN/credentials or coerced during cash-out	User education on credential safety; community awareness messaging; safe cash-out guidance; flag suspicious patterns
Liquidity / cash-out constraints: If cash-out is needed, insufficient liquidity or agent issues create delays	Select reliable cash-out points; agree service standards; monitor liquidity indicators; stagger payouts if needed; have backup cash-out locations
Delays due to unclear responsibilities or	Simple coordination cadence (weekly ops, monthly

slow decision-making	steering); named focal points; action tracker;
Reputational risk: Any incident (delays, disputes, perceived unfairness) reduces trust in the program	Proactive comms; transparent reporting; clear grievance/dispute process via community governance; rapid corrective actions; document lessons learned
Sustainability/Scale: The Pilot cannot be sustained after the initial funding ends	Early cost model for scale; identify long-term owner/operator; document efficiency gains; prepare scale roadmap and partnership plan

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)	Medium Term (6–18 months)	Long Term (18+ months)
From Architecture to First Verified Payout *Integrating Tech Phases 1, 2, & 3* **1. Design & Governance Alignment (Weeks 1-6):** • Define custody and multisig models aligned with *Usos y Costumbres*. • Finalize threat modeling (PIN reset, agent collusion risks). • **Deliverable:** On-chain specifications & UX Wireframes. **2. Core Implementation (Weeks 7-20):** • **Backend:** Implement custodied wallet service, PIN hashing, and	**Live Operations & Institutionalization** *Integrating Tech Phase 4* **1. Live Pilot Operation:** • Run multiple payout cycles to stress-test the "Hybrid Rail" (Cash-out vs. Bank Transfer). • **Gender Analysis:** Monitor women's usage rates vs. men to validate SDG 5 impact. • **System Tuning:** Refine UX based on feedback (e.g., simplifying PIN entry, offline queuing). **2. Operational Handover:** • Formalize reconciliation routines between Carbon Plus finance team and the blockchain ledger.	**Systemic Replication & Ecosystem Growth** **1. The "Forest-Chain" Kit:** • Package the solution (Tech + Governance Templates) for rapid deployment to other **Carbon Plus projects** (13,000+ landowners). **2. Broader Use Cases:** • Adapt the rail for other payment types: Ecosystem Service Payments, Shock-Responsive Transfers, or Government Social Benefits. **3. Financial Sustainability:** • Transition from "Pilot Funding" to a sustainable

reserve-backed token issuance. • **Frontend:** Build Landholder App (Balance/History) and Admin Dashboard for Comisariado. • **Integration:** Connect to Cardano Mainnet (via Blockfrost). **3. Deployment & "Genesis" Payout (Weeks 21-26):** • Onboard pilot cohort (**50 users**, 50% women) and 2-3 **Yastás** agents. • Execute first "Dry Run" followed by the first real-money payout. • Train "Digital Ambassadors" for local support. **Key Delvierables** • **Production-Grade MVP:** Fully deployed on Mainnet. • **Governance Config:** Smart contracts hard-coded with Assembly distribution rules. • **First Audit Log:** Immutable on-chain record of the first payout cycle. • **Trained Agents:** Local liquidity points operational.	• Establish "Dispute Resolution" protocols for lost PINs or agent errors. **3. Expansion:** • Scale from 50 users to the full **4,500 landowner base**. • Onboard additional agent networks to increase liquidity coverage. **Deliverables** • **Evidence Report:** Data on transaction speed, cost savings, and gender inclusion. • **Standard Operating Procedures (SOPs):** Manuals for Admin, Agents, and Users. • **Compliance Validation:** Report proving alignment with CAR Assurance Fund requirements.	**SaaS / Service Fee model** covered by project administration costs. • **Replication Kit:** Open-source or licensed software package. • **Regional Hub:** San Miguel Chimalapa established as the center of excellence for digital climate finance. • **New Verticals:** Integration with renewable energy or agroforestry projects.

5.9.2 Budget for Phased Implementation (post-acceleration)

Short Term (0–6 months) – Pilot implementation: US$173,600

Covers: High-intensity technical build (Architecture, MVP, Cardano Mainnet integration), Project Management, hardware (Starlink/Routers), and the pilot evaluation for the initial 50-person cohort.

Medium Term (6–18 months) – Scale within programme (4,500 users): ~US$150k–250k

Covers: Operational scaling to the full 4,500 landowner base. Note: *While heavy development costs decrease, resources shift toward field operations,* including mass onboarding brigades, extended agent network training, and formalizing dispute resolution protocols.

Long Term (18+ months) – Replication to new communities/sites: ~US$80k–150k per additional site

Covers: Deployment of the "Forest-Chain" Replication Kit. Cost is driven primarily by governance adaptation (legal alignment with new *Usos y Costumbres*), local stakeholder training, and configuration of the SaaS dashboard for new operators.

5.9.3 Financial Sustainability Model

How will the solution generate income post-pilot? Pricing model (e.g. subscriptions, licensing, transaction fees) and when it kicks in.

How will the solution generate income post-pilot?

Post-pilot, the solution will shift from grant-funded development to a cost-recovery service model paid by the programme/operator (Carbon Plus), ensuring that rightsholders can always use the solution free of charge.

The revenue model is driven by Risk Reduction Economics. By providing a 100% immutable audit trail, the solution lowers the project's "Social Risk" profile, allowing the developer to negotiate a lower "Risk Buffer" with the registry (Climate Action Reserve). The value of the carbon credits unlocked by this lower risk rating significantly outweighs the cost of the platform fees.

Revenue Streams: Income will be generated through three complementary streams:

1. **Programme Administration Fee:** A fee linked to each disbursement cycle (e.g., % of funds processed) to cover ongoing operations, reporting, and support. This is offset for the operator by the savings in manual reconciliation and audit costs.
2. **Operator Subscription:** A recurring fee for the "Transparency Dashboard," hosting, and maintenance (tiered based on volume and service levels).
3. **Replication & Licensing:** A one-time deployment fee for replicating the "Forest-Chain" kit to new communities or programmes (e.g., US$10,000 per site plus maintenance).

Timeline:

- **Months 0–6 (Pilot):** Grant-funded validation (Alliance + Accelerator).
- **Months 6–18 (Scale):** The cost-recovery model kicks in. Subscriptions and admin fees are introduced as the solution scales to the full 4,500 landowner base.
- **Month 18+ (Replication):** Licensing fees are applied as the solution is deployed to other Carbon Plus sites and external projects.

5.9.4 Partnership and Co-Financing Commitments
Which partners will be supporting the project (technical, financial, institutional) and what would be their contributions?

The project will be supported through a combination of technical, institutional, and financial/in-kind contributions from key partners. **Carbon Plus** will serve as the programme anchor, providing operational leadership, access to existing payout processes and beneficiary information, and coordination with community governance structures (primarily in-kind, with potential cost-sharing for recurring operating costs post-pilot).

The **San Miguel Chimalapa** community governance bodies (e.g., Asamblea/Comisariado) will provide legitimacy and oversight, validate beneficiary eligibility, support dispute resolution, and lead community mobilization and communications (in-kind time and governance processes).

The **Solution Maker/developers** (Vaka Consulting and technical partners) will deliver the platform build, configuration, and maintenance, financed initially through the pilot budget (financial) and potentially discounted or modular pricing for replication.

The **Better Than Cash Alliance and UNDP** will provide advisory support on responsible digital payments, inclusion and gender safeguards, partner coordination, and documentation/learning for scale (in-kind technical assistance).

In addition, relevant public-sector counterparts (e.g., SHCP) may contribute by supporting financial literacy, enabling conditions, and alignment with climate-finance governance (institutional support and in-kind coordination). They may be engaged for potential co-financing during the scale phase.

5.9.5 Fundraising Plan

Plan for securing funding through grants, ecosystem funds, investors or other sources.
Targeted sources and fundraising timeline.

The fundraising strategy will secure phased financing to move the solution from pilot to scale while transitioning to a sustainable cost-recovery model. In the first 0–3 months, partners will finalize a grant-ready package, including a pitch deck, and prepare a concise funder brief focused on transparency, inclusion, and climate/nature finance accountability.

Between months 3–6, the team will submit grants applications to fund pilot implementation and evaluation, while pursuing co-financing and in-kind contributions from ecosystem partners to demonstrate shared commitment.

From months 6–12, pilot evidence (monthly monitoring briefs and early results) will be used to secure scale funding through donor and ecosystem funds focused on digital public infrastructure, inclusive finance, governance, and climate/nature action, and to negotiate an operator service agreement to gradually absorb recurring operating costs.

Beyond 12–18 months, the team will pursue replication financing through multi-year donor programmes and portfolio-style grants to expand to additional communities, complemented by partnerships with relevant public institutions and private ecosystem actors, ensuring that replication is funded upfront while ongoing operations are covered through subscriptions and/or a small programme administration fee.

5.11 Legal & Compliance Considerations

Summary of key legal, regulatory, or compliance considerations.

Asset Classification: *The "Carbon-Peso" token is designed as a* ***closed-loop digital voucher*** *(representing a confirmed debt to the landowner) rather than a speculative cryptocurrency. This intends to keep the operation outside the strict purview of the "Virtual Assets" restrictions imposed on financial institutions by Banxico.*

Restriction Management: *To comply with Circular 4/2019, the system does not facilitate open-market trading of the token. It serves strictly as an internal accounting tool for disbursement.*

Legal "Win": *The project leverages the* ***National Code of Civil and Family Procedures****, which explicitly recognizes blockchain records as valid legal evidence.*

Compliance Application: *The "Audit Trail" generated by the platform serves as* ***"Prueba Plena" (Full Proof)*** *that the Comisariado fulfilled its fiduciary duty. This protects community leaders from legal liability and disputes regarding fund mismanagement.*

Anti-Money Laundering (AML/PLD) & KYC

- **Alignment:** The project complies with the Federal Law for the Prevention and Identification of Operations with Resources of Illicit Origin (LFPIORPI).
- **Implementation**: Although beneficiaries are low-risk, the platform enforces KYC by validating users against the Official Community Census.

Assurance Fund: The 100% on-chain traceability directly mitigates the "Financial Risk" and "Social Risk" categories in the CAR protocols. By proving funds reach the landowners, the project **maintains the eligibility** of the Carbon Plus verified asset (CAR 1963) and potentially reduces future risk buffer contributions (currently ~29%).

Balance Protection: To address community concerns (as noted in testing), the public ledger records transaction amounts but **masks beneficiary identities** using alphanumeric wallet addresses. Real-world identities are held off-chain in a secure, permissioned database accessible only to the Comisariado for audit reconciliation.

5.12 Visibility Assets / Communication Materials

1–2 slides, visuals, or storylines for use by UNDP or partners to promote your solution.

Project summary: The Alliance and Carbon Plus México joined UNDP's SDG Blockchain Accelerator to tackle inefficiencies in managing US$3.25M in carbon-credit payouts for 4,500 community members in Oaxaca. The project will pilot a blockchain-based system ensuring traceable, transparent, and inclusive payments, addressing current issues of opacity, delays, and exclusion.

Main storyline: By digitizing a $3.25M cash flow with a digital payments system and Blockchain technology, community-approved governance, and inclusive rails, this project transforms risk into trust—proving impact, protecting certification, and ensuring every beneficiary gets paid securely and transparently.

Visuals here:
https://drive.google.com/file/d/1dSeLnroxZoxkjEADnc5AXVk5bAlgXANb/view?usp=drive_link

Social media messages here:
https://docs.google.com/document/d/1ICCOUJevKfnbl_5198EWQrXrsMICP8uV/edit?usp=drive_link&ouid=114743703732004844390&rtpof=true&sd=true

Additional Pictures:
https://drive.google.com/drive/folders/1ajvC8_r7Ypf0A4WJ1ND4yC3jlEMRoNB7?usp=sharing

5.13 Pitch Deck

Concise and compelling presentation for stakeholders, funders, and partners. ((Please upload to your google drive folder)

The Pitch Deck is available here.

5.14 Final Demo Presentation

An engaging walkthrough of the live solution and its implementation roadmap at the end of the program.

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

Deliverable	Target Completion (Week # or Date)
1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	Done
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	Done
3. Solution Overview (core features, tech stack, blockchain functionality)	Done
5.1 Solution Demo (PoC)	Done
5.2 Testing & Feedback Documentation	Done
5.3 Feedback Integration Summary	Done
5.4 Pilot-Ready Solution	Not applicable
5.5 Confirmed Pilot Partnership(s)	Done
5.6 Stakeholder Alignment Matrix	Done
5.7 Budget Proposal (Pilot)	Done
5.8 Impact Measurement Framework (KPIs, SDG indicators, risk plan)	Done
5.9 Sustainability Plan & Post-Acceleration Roadmap (implementation timeline, budget, partners, fundraising)	Done
5.11 Legal & Compliance Mapping	Done

Deliverable	Target Completion (Week # or Date)
1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	Done
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	Done
3. Solution Overview (core features, tech stack, blockchain functionality)	Done
5.1 Solution Demo (PoC)	Done
5.2 Testing & Feedback Documentation	Done
5.12 Visibility Assets / Communication Materials	Done
5.13 Pitch Deck	Done